

CW3 - Forces & Motion w/ Newton ①

The modeling work that we do in classical mechanics leads us to EOMs. These

EOMs can be investigated in a number of

ways: (1) finding trajectories $\rightarrow x(t), v(t)$

later $\rightarrow x(v)$ & $x(p)$
(phase trajectories)

(2) creating phase space diagrams

$x(v)$ or $x(p)$ for a region
of x & v (or p)

(3) fixed points and stability analyses

$\dot{\vec{x}} = 0$ gives $\vec{x}_*$ critical pts.

and so on....

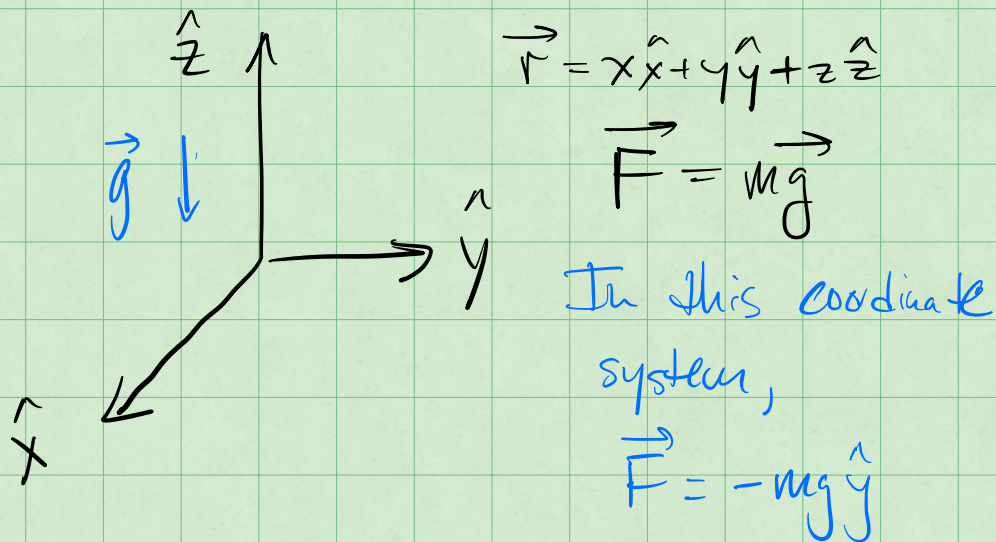
We will start with the:

FBD $\rightarrow$ EDM $\rightarrow$ trajectory pipeline

Throughout the analyses that we do, we will ask conceptual questions about these systems.

While doing that we will try also to make (2)
 clear a number of processes that help us
make sense of new models.

Example: Falling Object



Focus on the 1D problem

$\vec{F}_{\text{Earth}} = -mg\hat{y} = m\vec{a} = m\ddot{\vec{r}}$
 $\ddot{\vec{r}} = -g\hat{y}$
 $\ddot{y} = \dot{v} = -g$
 $v(t) - v_0 = \int_{v_0}^{v(t)} dv = -g \int_{t_0}^t dt' = -g(t - t_0)$

$$\boxed{v(t) = v_0 - g(t - t_0) \quad \text{trajectory of } v} \quad (3)$$

Integrate again to find $y(t)$,

$$\boxed{y(t) = y_0 + v_0 t - \frac{1}{2} g t^2 \quad \text{trajectory of } y}$$

Let's add drag to the model

Drag is the result of collisions with the falling body. Drag models are empirically developed as the specifics of how those collisions impact the moving body are quite complex. The two simplest models we have for drag are:

Linear Drag: $F_{\text{lin}} = \gamma v \rightarrow \begin{matrix} \text{small?} \\ \text{slow} \end{matrix}$

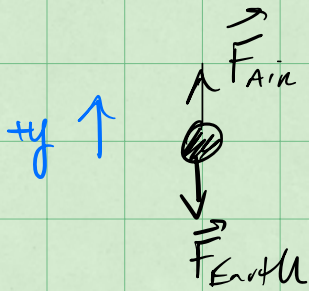
Quadratic Drag: $F_{\text{quad}} = D v^2 \rightarrow \begin{matrix} \text{large?} \\ \text{fast} \end{matrix}$

Both point opposite the velocity, $\vec{a} = -\frac{\vec{v}}{|\vec{v}|}$ 

Critically: you need to consider your coordinate system

Example: Quadratic Drag (1D)

(4)



$$\vec{F}_{\text{net}} = \vec{F}_{\text{AIR}} + \vec{F}_{\text{EARTH}} = m \vec{a} = m \vec{r}''$$

$$F_{\text{AIR}} = Dv^2 \quad F_{\text{EARTH}} = mg$$

$$m\ddot{y} = -mg + Dv^2$$

EOM:

$$\ddot{y} = -g + \frac{D}{m} v^2$$

let $\tilde{D} = D/m$

How do we solve this EOM?

- Separation of variables,
we try to right each side as
functions only of v and t

$$\int f(v) dv = \int g(t) dt$$

If we can integrate them we will
get a closed form relationship between
 v and t .

$$\ddot{y} = -g + \tilde{D}v^2 \Rightarrow \dot{v} = \frac{dv}{dt} = -g + \tilde{D}v^2 \quad (5)$$

$$\frac{dv}{dt} = -g + \tilde{D}v^2$$

$$\frac{dv}{-g + \tilde{D}v^2} = dt$$

$$\frac{dv}{dt} = 0 = -g + \tilde{D}v^2 \Rightarrow v_t = \sqrt{g/\tilde{D}}$$

terminal velocity

$$\frac{dv}{g[(v/v_t)^2 - 1]} = dt$$

$$\text{so } f(v) = \frac{1}{(v/v_t)^2 - 1}$$

$$g(t) = g$$

$$\frac{dv}{(v/v_t)^2 - 1} = g dt$$

Drop from rest

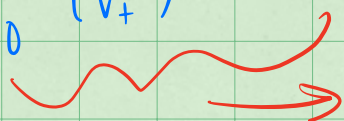
$$\int_0^{v(t)} \frac{dv'}{(v'/v_t)^2 - 1} = \int_0^t g dt'$$

a trajectory
requires
initial
conditions

$$v(0) = 0$$

$$\int_0^v \frac{dv'}{\left(\frac{v'}{v_+}\right)^2 - 1} = \int_0^t g dt$$

(6)



look up integrals

⇒ Wolfram Alpha, sympy, etc.

$$-v_+ \tanh^{-1}\left(\frac{v}{v_+}\right) = gt$$

$$v(t) = v_+ \tanh\left(\frac{-gt}{v_+}\right)$$

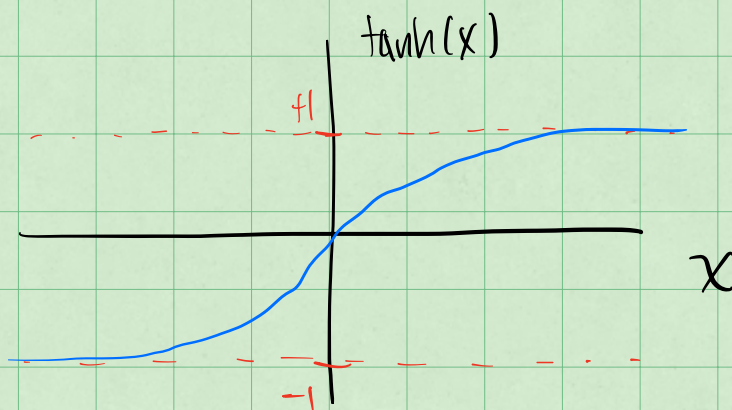


$$\text{where } v_+ = \sqrt{g/\alpha} = \sqrt{mg/D}$$

New equation! How do we check it?

⇒ units?

⇒ limits?



Example: 2D Quadratic Drag (7)

$$\vec{F}_D = -D \vec{v} |\vec{v}|$$

Need coordinate system.

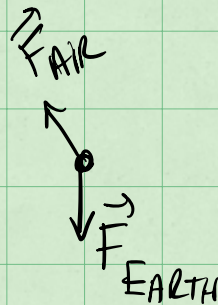
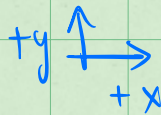


$$\vec{r} = x \hat{x} + y \hat{y} = \langle x, y \rangle$$

$$\vec{v} = v_x \hat{x} + v_y \hat{y} = \langle v_x, v_y \rangle$$

$$\vec{a} = a_x \hat{x} + a_y \hat{y} = \langle a_x, a_y \rangle$$

FBD



$$|\vec{v}| = \sqrt{v_x^2 + v_y^2}$$

Newton 2nd

$$ma_x = F_{Air, x}$$

$$ma_y = F_{Air, y} - F_{EARTH}$$

$$F_{Air, x} = D |\vec{v}| v_x$$

$$F_{Air, y} = D |\vec{v}| v_y$$

$$F_{EARTH} = mg$$

$$ma_x = m\ddot{x} = Dv_x \sqrt{v_x^2 + v_y^2}$$

(8)

$$ma_y = m\ddot{y} = Dv_y \sqrt{v_x^2 + v_y^2} - mg$$

Coupled EoMs (cannot go further)

$$\dot{v}_x = \tilde{D} v_x \sqrt{v_x^2 + v_y^2}$$

with $\tilde{D} = D/m$

$$\dot{v}_y = \tilde{D} v_y \sqrt{v_x^2 + v_y^2} - g$$

We need another approach since there's no analytical solution to these Differential Equations

⇒ cannot form,

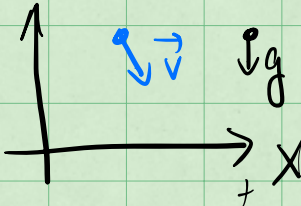
$$f_1(v_x) dv_x = g_1(t) dt \quad \text{independent}$$

$$f_2(v_y) dv_y = g_2(t) dt \quad \text{EoMs}$$

so separation of variables is not possible

Example: Linear Drag in 2D

⑨

$$\vec{F}_{\text{Lin}} = -m\gamma \vec{v} + y \uparrow$$


$$\vec{F}_{\text{net}} = \vec{F}_{\text{Lin}} + \vec{F}_{\text{EARTH}}$$

$$ma_x = -m\gamma v_x$$

$$ma_y = -m\gamma v_y - mg$$

note v_x & v_y
can be +, -, or, 0.

they are components

EOMs

$$\dot{v}_x = -\gamma v_x \quad \dot{v}_y = -\gamma v_y - g$$

These are decoupled so we will try
separation of variables

$$\dot{v}_x = \frac{dv_x}{dt} = -\gamma v_x$$

$$\frac{dv_x}{v_x} = -\gamma dt$$

Integrate!

$$\int_{v_{0x}}^{v_x} \frac{dv_x'}{v_x'} = -\gamma \int_{t=0}^t dt'$$

(10)

$$\ln(v_x/v_{0x}) = -\gamma t$$

$$v_x(t) = v_{0x} e^{-\gamma t}$$

trajectory
for v_x

$$\frac{dx}{dt} = v_x$$

$$\int_{x_0}^x dx' = \int_{t=0}^t v_{0x} e^{-\gamma t'} dt'$$

$$x - x_0 = -\frac{v_{0x}}{\gamma} (e^{-\gamma t'})_0^t$$

$$x - x_0 = -\frac{v_{0x}}{\gamma} (e^{-\gamma t} - e^0)$$

$$x(t) = x_0 + \frac{1}{\gamma} v_{0x} (1 - e^{-\gamma t})$$

trajectory
for x

$$\dot{v}_y = -\gamma v_y - g$$

(11)

$$\frac{dv_y}{dt} = -\gamma v_y - g$$

$$\frac{dv_y}{\gamma v_y + g} = -dt \Rightarrow \frac{dv_y}{v_y + g/\gamma} = -\gamma dt$$

Integrate!!!
 $t=0, v_y = v_{0y}$

$$\int \frac{dx}{a+x} = \ln(a+x) + C$$

$$\int_{v_{0y}}^{v_y} \frac{dv_y'}{v_y' + g/\gamma} = \ln \left(\frac{g/\gamma + v_y}{g/\gamma + v_{0y}} \right) = -\gamma t$$

$$\left(\frac{g}{\gamma} + v_y \right) = \left(\frac{g}{\gamma} + v_{0y} \right) e^{-\gamma t}$$

$$v_y(t) = \frac{g}{\gamma} (e^{-\gamma t} - 1) + v_{0y} e^{-\gamma t}$$

trajectory for v_y .

$$\frac{dy}{dt} = v_y$$

(12)

$$\int_{y_0}^y dy' = y - y_0 = \int_{t=0}^t \left\{ \frac{g}{\gamma} (e^{-\gamma t'} - 1) + v_{0y} e^{-\gamma t'} \right\} dt'$$

$$\begin{aligned} y - y_0 &= \frac{g}{\gamma} \left(-\frac{1}{\gamma} e^{-\gamma t'} - t' \right)_0^t + v_{0y} \left(-\frac{1}{\gamma} e^{-\gamma t'} \right)_0^t \\ &= \frac{g}{\gamma} \left(-\frac{1}{\gamma} e^{-\gamma t} - t + \frac{1}{\gamma} \right) + v_{0y} \left(-\frac{1}{\gamma} e^{-\gamma t} + \frac{1}{\gamma} \right) \end{aligned}$$

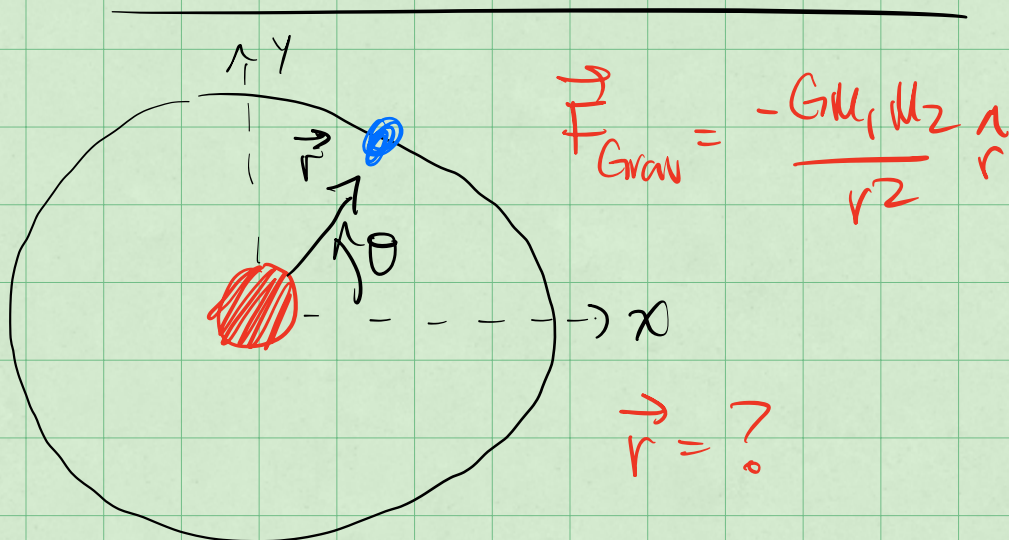
$$y(t) = y_0 + \frac{v_{0y}}{\gamma} - \frac{v_{0y}}{\gamma} e^{-\gamma t} - \frac{g}{\gamma} t + \frac{g}{\gamma^2} - \frac{g}{\gamma^2} e^{-\gamma t}$$

$$y(t) = y_0 - gt/\gamma + \frac{1}{\gamma} \left(v_{0y} + \frac{g}{\gamma} \right) (1 - e^{-\gamma t})$$

trajectory for y

Example: 2D Newtonian Gravitational Model (Sun - Earth)

(13)

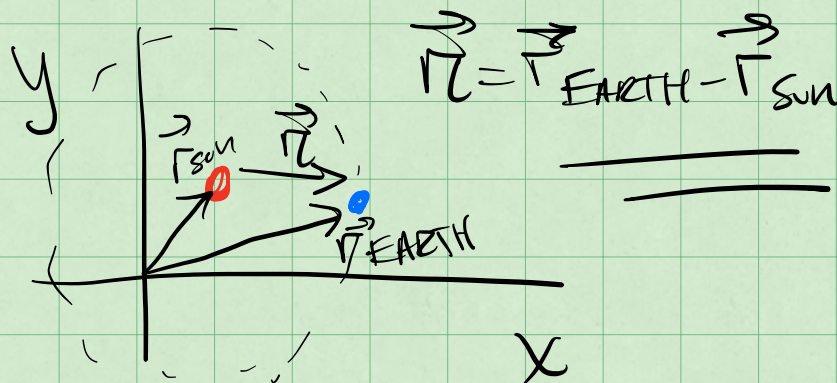


$$\vec{F}_{\text{Grav}} = -\frac{GM_1 M_2}{r^2} \frac{\vec{r}}{r}$$

$$\vec{r} = ?$$

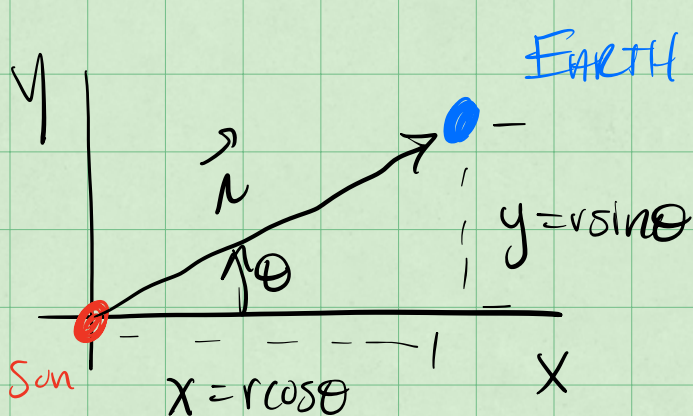
$$\vec{r} = \vec{r}_{\text{object 1}} - \vec{r}_{\text{object 2}}$$

move
Sun off
(0,0)



$$\vec{r} = \vec{r}_{\text{EARTH}} - \vec{r}_{\text{Sun}}$$

Back to the Simplified 2D Model



(14)

$$M_{\odot} = 2 \cdot 10^{30} \text{ kg}$$

$$M_E = 6 \cdot 10^{24} \text{ kg}$$

$$G = 6.67 \cdot 10^{-11} \frac{\text{N}}{\text{kg}^2 \text{ m}^2}$$

$$\vec{F}_{\text{grav}} = -G \frac{M_{\odot} M_E}{|\vec{r}|^3} \vec{r}$$

$$|\vec{r}| = \sqrt{x^2 + y^2}$$

$$= 1 \text{ A.U.}$$

$$= 1.5 \cdot 10^{11} \text{ m}$$

$$\vec{F}_{\text{NET}} = \vec{F}_{\text{grav}} = m \vec{a} = m \langle a_x, a_y \rangle$$

$$F_x = - \frac{G M_{\odot} M_E x}{|\vec{r}|^3}$$

$$F_y = - \frac{G M_{\odot} M_E y}{|\vec{r}|^3}$$

$$a_x = \dot{v}_x = \ddot{x} = \frac{-G M_{\odot} x}{(x^2 + y^2)^{3/2}}$$

$$a_y = \dot{v}_y = \ddot{y} = \frac{-G M_{\odot} y}{(x^2 + y^2)^{3/2}}$$

EQNS
for Grav
(coupled)

Trajectories?

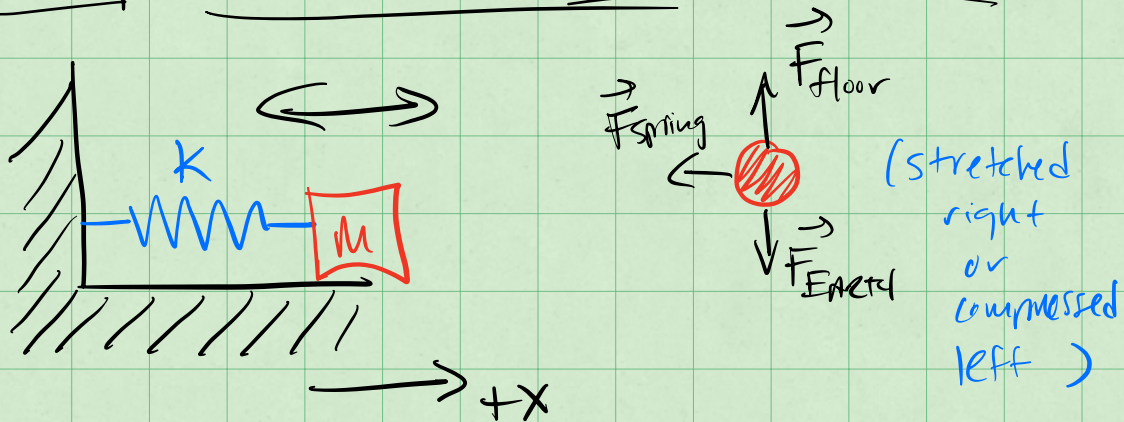
(15)

(1) Integrate EoMs

(2) Solve coupled EoMs → techniques for this

(3) solve numerically

Example: Simple Harmonic Oscillator (1D)



$$F_s = ks \leftarrow \text{stretch } (x - L_0)$$

relaxed length ↑

→ if we measure displacement from equilibrium (L_0) then $F_s = kx$ direction opposite of stretch

$$F_x = m a_x = m \ddot{v}_x = m \ddot{x} = -Kx$$

$$m \ddot{x} = -kx \quad \text{or} \quad \ddot{x} = -\frac{k}{m} x$$

EOM of the SHO (Very important result)

Solutions?

$$\frac{d^2 x}{dt^2} = -\frac{k}{m} x = -\omega^2 x \quad \text{where } \omega^2 = \frac{k}{m}$$

Try $x(t) = Ce^{i\omega t}$

$$\dot{x}(t) = i\omega Ce^{i\omega t}$$

$$\ddot{x}(t) = -\omega^2 Ce^{i\omega t} = -\omega^2 x(t) !!$$

$x(t) = Ce^{i\omega t}$ where $C = a + ib$ works!

So, maybe $\cos(\omega t) + \sin(\omega t)$?

Try $x(t) = A \cos(\omega t) + B \sin(\omega t)$

$$\dot{x}(t) = -\omega A \sin(\omega t) + \omega B \cos(\omega t)$$

$$\begin{aligned} \ddot{x}(t) &= -\omega^2 A \cos(\omega t) - \omega^2 B \sin(\omega t) \\ &= -\omega^2 x(t) !! \end{aligned}$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t) \text{ with } \omega^2 = k/m$$

generic trajectory for SHO

(7)